

1. What is a decision boundary, and how does it relate to classification tasks?

Answer : A decision boundary is a line or hyperplane that separates data into different states, these states help in classification.

2. How does the bias term in a neuron affect the position of the decision boundary?

Answer : The bias term in a neuron acts as an offset, it makes the decision boundary to shift away from the origin, without it the model can not classify the data that isn't centered around the origin.

3. Explain with an example how changing weights affects the orientation of the decision boundary.

Answer : The weights influence the slope and orientation angle of the decision boundary. As example , a dataset of study hour as x_1 and attendance as x_2 , here :

Scene (1) : if w_1 and w_2 both are 1 then we get a 45 deg boundary line . It means both feature has equal influence on the prediction result.

Scene (2) : if we take w_1 as 10 and keep w_2 as it is , the boundary will rotate and be much steeper. It means the model is now influenced mostly by the feature study hour , resulting in the classification biased.

4. Why is it important to visualize the decision boundary when training a model?

Answer : The decision boundary can clearly show if a model is overfitting, underfitting, it shows class separation, helps to verify learning that how the model updates weights and bias. How it is affecting the boundary to flip or rotate .

Math :

1. Weight Update (Perceptron Learning Rule)

A perceptron has:

- Initial weights: $w = [0.2, -0.1]$
- Bias: $b = 0.1$
- Learning rate: $\eta = 0.1$

For a training example with:

- $x = [1, 1]$
- Target $t = 1$

If the perceptron output is $y = 0$, calculate the **new weights and bias** after one update using:

$$w_{\text{new}} = w + \eta \cdot (t - y) \cdot x$$

$$b_{\text{new}} = b + \eta \cdot (t - y)$$

Answer : Given,

Initial weights $w = [0.2, -0.1]$

Bias $b = 0.1$

Learning rate $\eta = 0.1$

Input $x = [1, 1]$

Target $t = 1$

Current output $y = 0$

The weight update formula is $w_{\text{new}} = w_{\text{old}} + \eta \cdot (t - y) \cdot x_i$

After putting the values we get $w_1(\text{new}) = 0.2 + 0.1 = 0.3$ and $w_2(\text{new}) = -0.1 + 0.1 = 0.0$

The new bias update formula is $b_{\text{new}} = b_{\text{old}} + \eta \cdot (t - y)$

So new bias $b = 0.1 + 0.1 = 0.2$